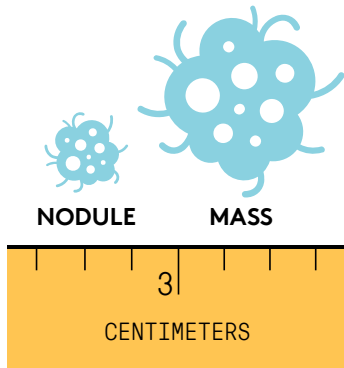
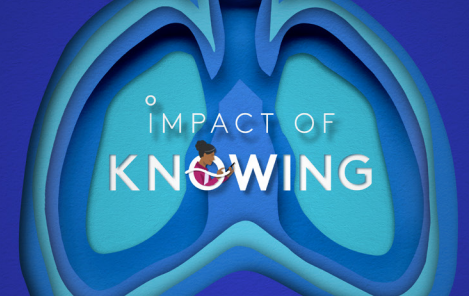


All About Lung Nodules



What are lung nodules?

American Lung Association defines a lung nodule as a small mass of dense tissue on the lung.¹ According to Dr. Lam at the Cleveland Clinic, a nodule is generally less than 3 cm in size. Lung masses, or larger growths, are more than 3 cm in size. Only three or four out of 100 nodules are cancerous, or less than 5%.² They are often discovered during a chest X-ray or CT scan performed for another reason.

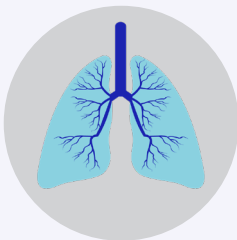
Potential causes

Benign (noncancerous) lung nodules can be caused by a variety of factors, including respiratory inflammation, infection, and scarring.³ The chances of a nodule being cancerous are increased if:⁴

- You are over the age of 65
- You have a history of smoking
- You have a family history of lung cancer
- Received radiation therapy to the chest
- Were exposed to asbestos, radon or secondhand smoke

When are lung nodules found?¹

Lung nodules are found during imaging tests, such as a CT scan or a chest X-ray. You could have one of these scans for several reasons:



As part of a lung cancer screening for patients at high risk



To determine the cause of symptoms you are experiencing, such as an unexplained cough



For an unrelated health condition that requires imaging of the lungs or a nearby area

Nodule growth and size⁵

Nodules that are over three centimeters in size are considered a mass. A mass of this size with irregular borders is more likely to be cancerous. Cancerous nodules also grow more quickly than benign nodules, and can double in size every four months, but sometimes as quickly as 25 days and some as slowly as 15 months. Non-cancerous nodules grow slowly, if at all.

If your nodule has not grown for two years after discovery, you should talk to your doctor about if you are able to stop monitoring it. You should also discuss biopsy options with your doctor to determine whether the nodule is cancerous or not.



Finding a nodule¹

Once a nodule is discovered, you will most likely be referred to a pulmonologist (a doctor who specializes in diseases of the respiratory system). Your doctor will discuss their findings and potential next steps based on the size, shape and location of the nodule. You may need additional tests, such as:

Blood tests

To rule out other conditions, such as tuberculosis.

PET (positron emission tomography) scan

To get a more detailed look at the nodule. The size and shape may provide more information on the likelihood it is cancerous or noncancerous.

Biopsy

A test in which a small amount of tissue from the nodule is removed to be examined closely under a microscope for cancerous cells. This is the only way to definitively diagnose cancer.

What is a lung biopsy? When should I have one?⁶

A lung biopsy is a procedure where your doctor will obtain a small sample of lung tissue from the nodule and look at it under a microscope to see if cancer or another disease is present.

Your doctor will consider several factors when deciding if you should have a lung biopsy. Your risk factors for lung cancer, as well as the size and shape of the nodule, will be key. If the nodule is more than a half an inch or has irregular borders, you should talk to your doctor about a biopsy.



To learn about minimally invasive options like robotic-assisted lung surgery using da Vinci technology, visit <https://www.intuitive.com/en-us/patients/procedures/thoracic/lung-surgery>.

The information provided is not intended for medical diagnosis or treatment or as a substitute for professional medical advice. Consult with a physician or qualified HCP for appropriate medical advice.

¹ [Lung Nodules \(Pulmonary Nodules\)](#). American Lung Association. 25 January 2024.

² [What Are the Chances a Lung Nodule or Spot Is Cancer?](#) Cleveland Clinic. 25 January 2024.

³ [Benign Lung Tumor](#). Diseases & Conditions. Cleveland Clinic. 23 January 2024.

⁴ [Pulmonary Nodules](#). Diseases & Conditions. Cleveland Clinic. 6 February 2024.

⁵ [Pulmonary Nodules](#). University of Rochester Medical Center. Web. 24 January 2024.

⁶ [Lung Biopsy](#). Johns Hopkins Medicine. 25 January 2024.

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